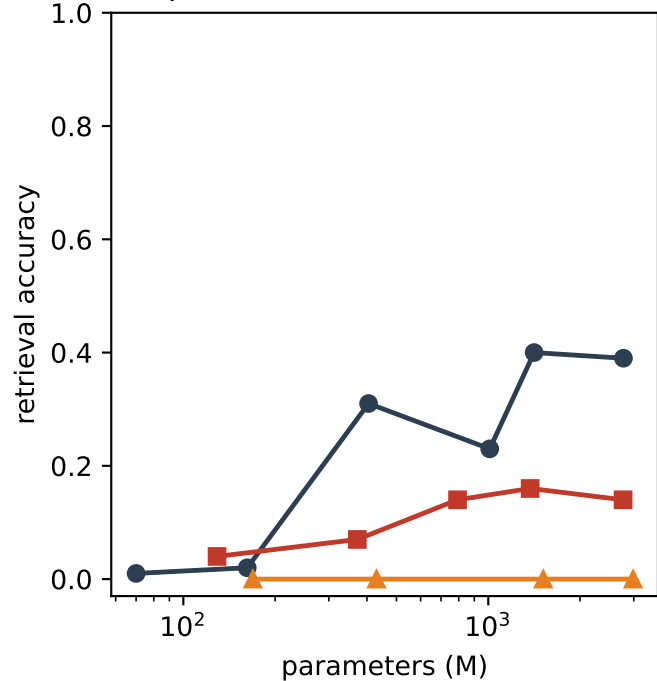
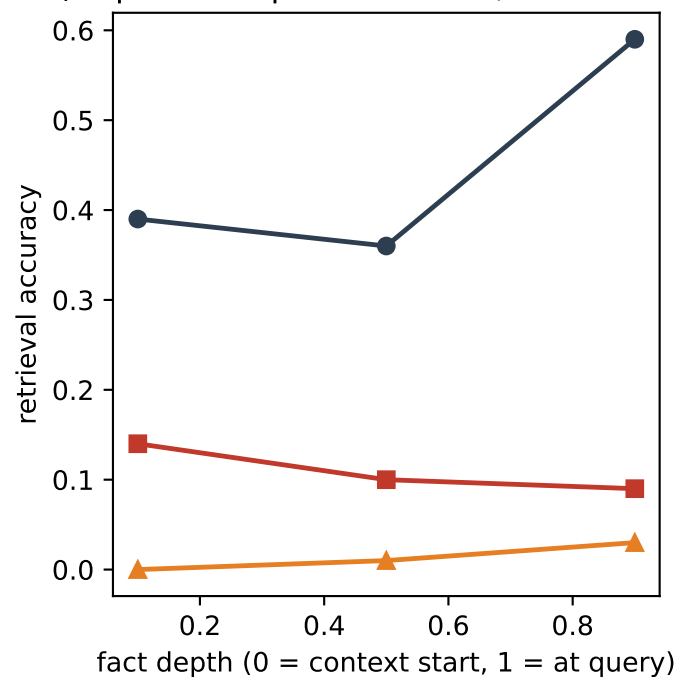


Exp A: representational convergence $\neq$ capability convergence (Pythia / Mamba / RWKV, all trained on The Pile)

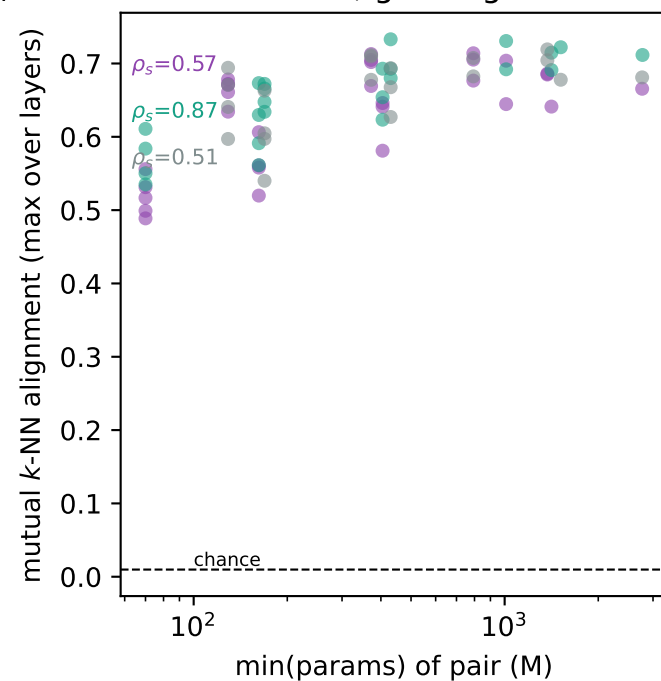
(a) capability stratifies by access structure
($N = 16$ facts, depth 0.1, $L = 896$, same Pile data+tokensize)



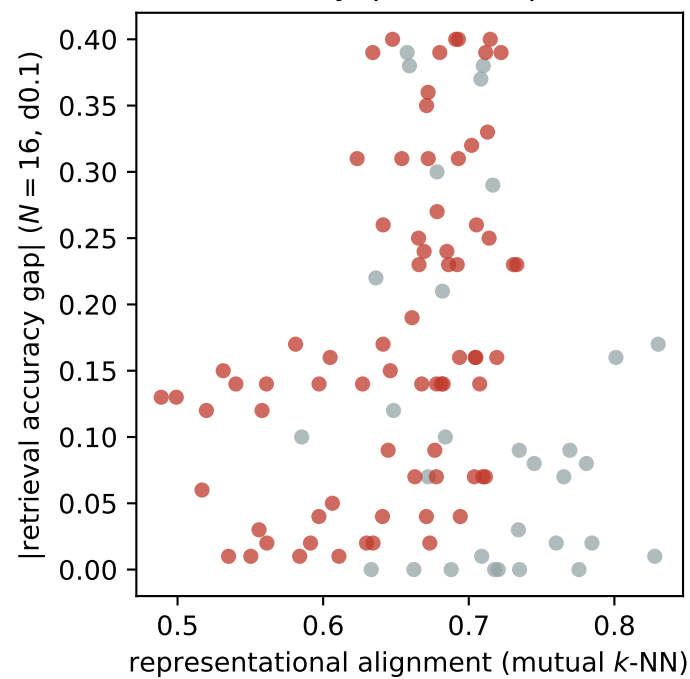
(b) recency signature at $\sim 3B$
(size = compressive state, flat = index)



(c) PRH replication: representations align across families, growing with scale



(d) alignment does not buy retrieval cross-family: $\rho_s = 0.48$ ($p = 0.00$)



Pythia (attention) Mamba (pure SSM) RWKV-4 (linear RNN) cross-family pair within-family pair